

Slot opening optimization of surface mounted permanent magnet motor for cogging torque reduction

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ABSTRACT

In this paper, slot opening skew method is used for cogging torque reduction. A three layer stator model is considered for a six-pole PM-BLDC motor (a PM-BLDC motor with 18-slots, six-poles and length of 5 cm) and then a 2D dual model is extracted for this 3D slot opening skew model. The angular shifts of slot opening position in the first and third layers than middle layer are considered as optimization parameters. Slot opening shape is optimized by using different optimization algorithms, such as, the response surface methodology (RSM), the genetic algorithm (GA) and the particle swarm optimization (PSO). In order to using of GA and PSO, the analytical relationship is derived for the air-gap permeance function. The optimization results of these algorithms are being consistent with each other and are verified with FEA results. The results show the significant reduction of cogging torque about 77%.

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1. Introduction

One of the important challenges in design of the PM electrical machines is to reduce the cogging torque which affects the machine starting performance and results in shaft vibration, acoustic noise and damage to drive component. The main causes of torque pulsation are as follows:

- Cogging torque.
- Harmonics of the back EMF.
- Saturation of magnetic circuit.
- Current harmonics.

Cogging torque is main cause of torque pulsations and is due to interaction between the air gap flux distribution and the slotted stator structure. There are different approaches to the cogging torque reduction from either machine design or control perspectives [1]. The machine design techniques minimize the cogging torque by optimization of the machine's structural parameters, whereas control techniques require the scholastic current excitations and are highly dependent on trustiness and accuracy of the used sensors. Therefore, the machine design techniques are more effective than the control techniques.

There are many methods for cogging torque reduction in surface-mounted permanent magnet motors, such as, without skewing

techniques [2–5], PM or slot skewing [6–10], notch insertion in stator teeth [11], the stator or rotor teeth-pairing [12–15], asymmetrical distribution of magnets [16], the conformal mapping method [17], the magnet segmentation method [18–27], application of dual structure [28], the stator asymmetry [29], the slot opening skew [30], the uneven magnetization and tooth curvature [31].

One of the most famous methods for cogging torque suppression is PM or slot skewing which are discussed in [6–10]. In practice, due to the end effect, the cogging torque cannot cancel completely by skewing. Skewing has many objections. It makes the motor construction is more complex, the leakage flux is increased and the output torque reduced particularly in motors with the small number of teeth per pole. In [9], the torque ripple is reduced using step-skewed magnet. Really, the magnet skewing technique is more efficient in motors with higher optimum skew angle. In [11], for compensation of the cogging torque by skewing, the notches are inserted in the stator teeth and the optimal number and shape of theses notches are obtained by optimization using the combined numerical and analytical methods.

Other techniques for the cogging torque reduction are stator or rotor teeth-pairing [12–15]. In this method, two different type of stator or rotor tooth width is used. Rotor teeth pairing is done circumferentially [12,13] or axially [15].

Despite other methods for cogging torque reduction, the using of magnet segmentation method makes that the motor structure is not more complex. In this method, the key idea is to set the distribution of the air-gap flux density by segmenting the magnet pole into several magnet blocks with radial magnetization. First,

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this method is proposed in [18]. This method is used in [19,20] to optimize the displacement parameters between PM segments, for the air gap flux density harmonics elimination [21,22], eddy current loss minimization in permanent magnets [23–26]. Also, in [27], a multi-objective optimization process has been proposed to reduce the cogging torque with minimum loss in the output torque in the segmented pole PMSMs based on the semi-analytical model with improved accuracy. The optimization of PM surface profile is applied to reduce the cogging torque in [32], so that the air-gap is non-uniform in radial direction and uniform in axial direction. Also, in [33], a pole shape optimization based on analytical model is presented to reduce the thrust ripple and back-emf harmonics.

A novel air-gap profile is proposed in [34] for SP PM-BLDC motors that lead to improve the starting torque and to reduce the cogging torque. In [35], Based on an improved subdomain model for predicting open circuit magnetic field of surface-mounted permanent magnet machines, cogging torque has calculated using Maxwell stress tensor.

The IPM configurations produce significant cogging torque, therefore, the different methods are used to decrease the cogging torque in IPM motors. For example, the cogging torque is reduced without skewing in [4] and with skewing the stator in [10]. In [36], the optimization of the slot gap width and PM thickness of a stator interior permanent magnet (SIPM) machine is analyzed by FEM and it is shown that the cogging torque is decreased when the slot gap width and PM thickness is reduced. Also, the various flux barrier designs in an IPMSM are presented in [37] and their geometries are optimized using RSM to minimize the cogging torque. The pole face rotor core using several notches is designed to reduce the cogging torque in [38]. Meanwhile, in the axial flux permanent magnet machines, the influence of magnet shape and displacement between both stator sides [39] and the effect of stator slot displacement and skewing [40] for the cogging torque reduction are investigated.

In this paper, two techniques are presented for slot skewing, i.e. the conventional slot skewing and the slot opening skewing. We concentrate on the latter method in this paper. Here, we consider a three layers stator model for a six-pole PM-BLDC motor and the angular shift of slot opening position in the first and third layers than middle layer are considered as the optimization parameters. The optimal value of each angular shift is obtained using different optimization algorithms, such as, RSM, GA and PSO. The results show the drastic reduction in cogging torque.

2. Slot opening skew

In permanent magnet motors, the cogging torque has a periodic waveform versus mechanical position of rotor. Therefore, in view of the frequency analysis it is necessary to reduce the magnitude of its harmonics for cogging torque reduction. In practice, the cogging torque can be reduced through PM or slot skewing [6–10].

Skew can be implemented in two forms: discrete or continuous. In this paper, the discrete-type skew is discussed as shown in Fig. 1. In other view, the slot skewing can be done in two forms, the conventional slot skewing [7,9–10] or the skewing of slot opening [30]. As seen in Fig. 2, in conventional slot skewing method, the slot and slot opening are skewed, simultaneously. The conventional slot skewing cause in decentralizing of winding function which is not favorable in PM-BLDC motors. This is one of the main disadvantages of the conventional slot skewing. Since, the slot opening skew cause in the smoothing of the air-gap reluctance variations that is main reason of the cogging torque, therefore, in this paper the slot opening skew method is discussed which is shown in Fig. 3 for a typical slot.

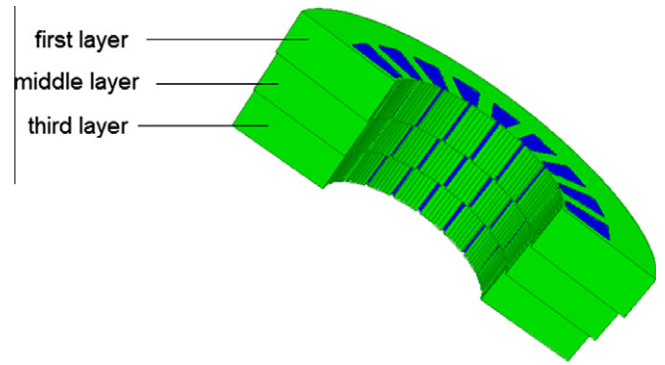


Fig. 1. Implementation of discrete-type skew.

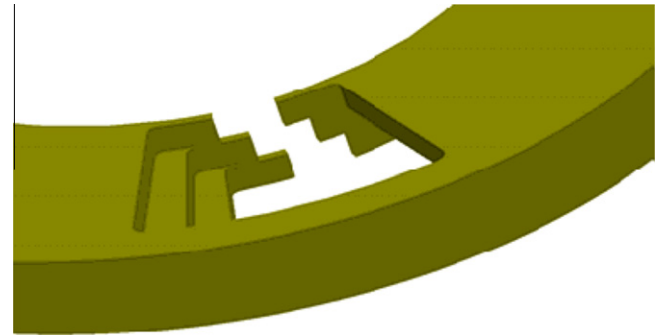


Fig. 2. The conventional slot skewing [30].

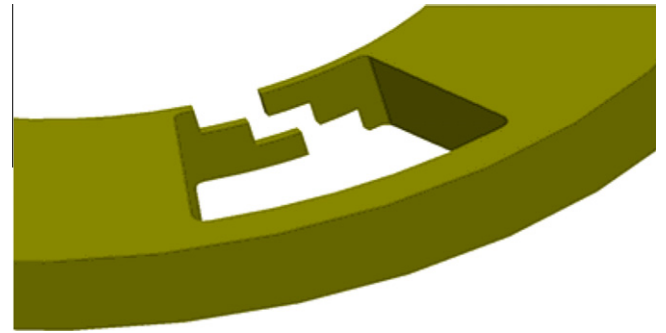


Fig. 3. Slot opening skew [30].

By using slot opening skew, the position of slot and conductor are not changed and consequently the winding function remain centralized and the trapezoidal waveform of back-emf of PM-BLDC motor is maintained against of the conventional slot skewing.

Implementation of conventional slot skewing and slot opening skewing for a three layer stator model of a six-pole PM-BLDC motor are shown in Figs. 2 and 3. As seen in Fig. 2, the first and third layers than middle layer has shifted by amount θ_1 and $-\theta_2$ angles. In contrary, Fig. 3 shows that only the slot opening of the first and third layers than middle layer has displaced. The view of each layer in Fig. 3 is shown in Fig. 4 independently.

If the stator layers are divided into number of poles, each of these divisions is called one block. Each block is placed under a pole. Since, the shape of each block and their positions than the magnet poles are identical, therefore, their cogging torque are similar. Consequently, total cogging torque is expressed as follows:

$$T_{cog} = P \cdot \sum_{l=1}^{\text{Number of layers}} \sum_{h=0}^{\infty} T_{hl} \cdot \sin(mh(\gamma + \beta_{hl})) \quad (1)$$

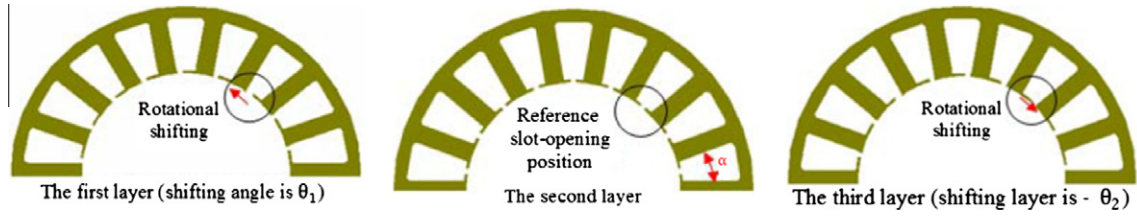


Fig. 4. The different layers of stator with shifted slot opening.

where T_{hl} is the fourier coefficient of cogging torque of each block in l th layer, m is the least common multiple of the number of slot and poles, P is number of poles, γ is the rotor position and β_{hl} is the phase shift in h th harmonic of cogging torque of a block in l th layer.

3. The dual structure for slot opening skew

In this part, a dual structure is derived for the model is shown in Fig. 3, so that the slot shape is uniform axially, but the teeth width distribution is non-uniform. The using of 2D FEA instead of 3D FEA is the main feature of this dual model and cause in the computing time reduction. To illustrate the derivation steps of this dual model, we consider a two layer stator model of a four-pole motor without consideration slot opening skew and with slot opening skew are shown in Figs. 5 and 6. As seen in Fig. 6, the shifting vector of slot openings is a three dimensional vector and the strip parts are tooth.

Since, the cogging torque is similar for the blocks that are placed under the same poles; we can replace the blocks N1, S1, N2 and S2 with N4, S4, N3 and S3, respectively. The new structure has shown in Fig. 7.

As seen in Fig. 7, the shifting vector is a two-dimensional vector. With applying this dual model to our favorite motor, the structure that is shown in Fig. 8 is obtained. Therefore, the dual structure has the distributed tooth width unevenly. In next part, it is shown that the non-uniform distribution of teeth width has unrespectable effect on the air-gap flux density.

In order to validate the 2D slot opening shift model, this dual model is compared with the 3D slot opening skew model that is shown in Fig. 9. The considered parameter for comparison is the cogging torque. In this 3D model, the slot opening in above and low layers are shifted than the middle layer $+2^\circ$ and -2° , respectively. For these conditions, the comparison result is shown in Fig. 10. As seen in this figure, there is quite conformity between the cogging torque wave shapes are obtained from these two models.

4. The simulation of skewing using FEM

In this part, both the conventional skewing and slot opening skew are applied to a three layer stator model of a six-pole PM-BLDC motor and their results are compared with each other. For simplification in comparison between the conventional skewing

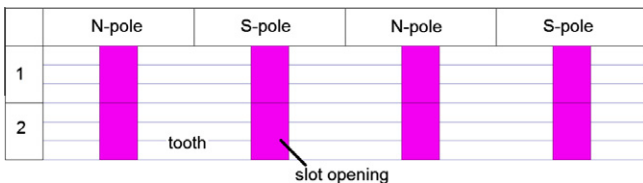


Fig. 5. The placement of slot openings without slot opening skew.

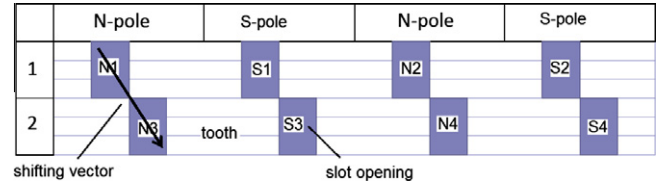


Fig. 6. The placement of slot openings with slot opening skew.

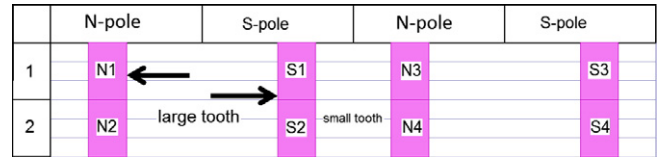


Fig. 7. The dual structure for slot opening skew in Fig. 6.

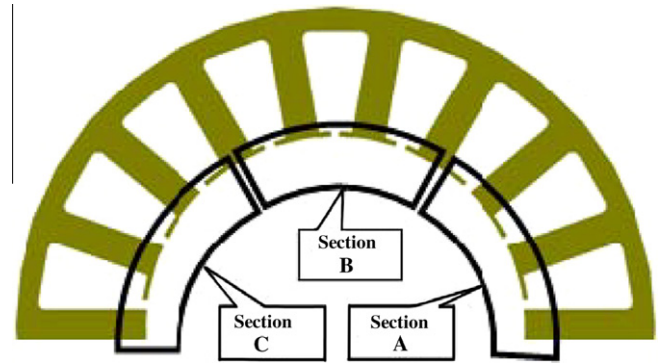


Fig. 8. The dual structure for our favorite motor.

and slot opening shift, in this section of paper we assume that the skew angles in the first and third layers than the second layer are θ and $-\theta$ (this assumption is applied for both skew-type)¹.

In conventional skewing method, the maximum allowable value of θ is equal to the half of slot pitch ($\theta_{\max} = 10^\circ$). In contrary, the maximum limit of θ is $\frac{\pi}{2}$ for the slot opening skew method that α is introduced in Fig. 4. The cogging torque results for both skewing methods are shown in Fig. 11. As seen in figure, the minimal value of cogging torque is similar for both skewing method, nearly.

The comparison of back-emf of phase-A for the both skewing methods is shown in Fig. 12 with considering the optimum skew angles are obtained for each skewing method. As seen in this figure, the back-emf wave form is sinusoidal nearly for the conventional skewing method, but, the trapezoidal wave shape of back-emf is maintained by using slot opening shift method. The frequency spectrum of back-emf wave forms is shown in Fig. 13a

¹ In optimization part, we assume that the angular shifts are not same in clockwise and counter-clockwise direction.

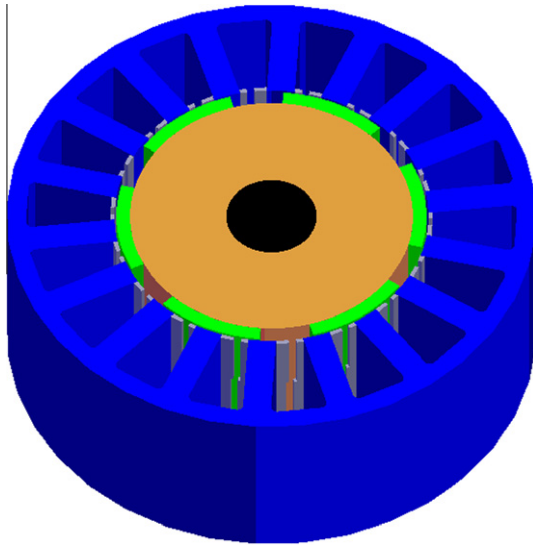


Fig. 9. The 3D slot opening skew model.

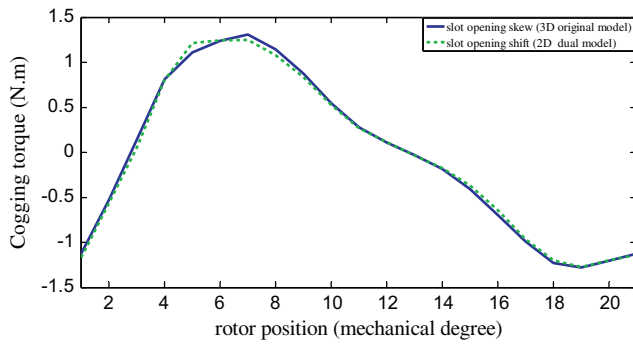


Fig. 10. The comparison of cogging torque between the 2D and 3D models of slot opening skew.

and b for the obtained results of the conventional skewing method and the slot opening shift method, respectively.

The THD² is calculated for the back-emf wave forms using following equation:

$$THD = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots + V_\infty^2}}{V_1} \quad (2)$$

The THD results are obtained from these two methods are shown in Table 1.

As seen, the THD is lower about 53% for the optimum back-emf is obtained from the conventional skewing method than the optimum back-emf is obtained from the slot opening shift method. Therefore, the obtained back-emf waveform of the conventional skewing is more sinusoidal.

Also, in Fig. 14, the influence of slot opening shift on the air gap flux density is shown with including the optimum shift angle is obtained for dual model³. As seen in this figure, the slot opening shift has an ignorable effect on the flux density distribution in air-gap.

² Total harmonic distortion.

³ In this figure, the flux density is drawn along a circular trajectory near to tooth.

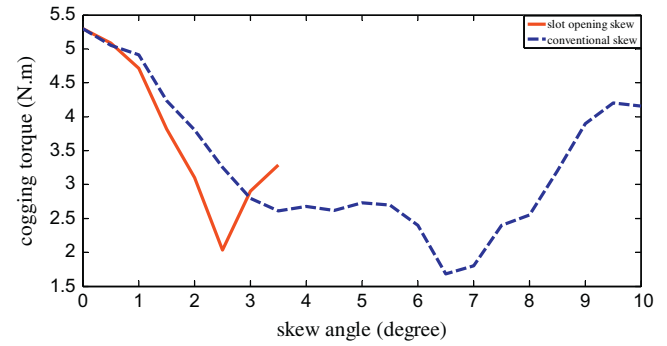


Fig. 11. The peak to peak value of cogging torque versus the skew angle for both skewing methods.

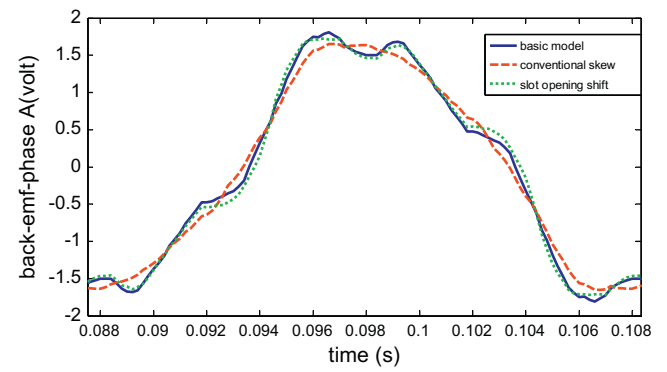


Fig. 12. The back-emf wave form for different slot opening shapes.

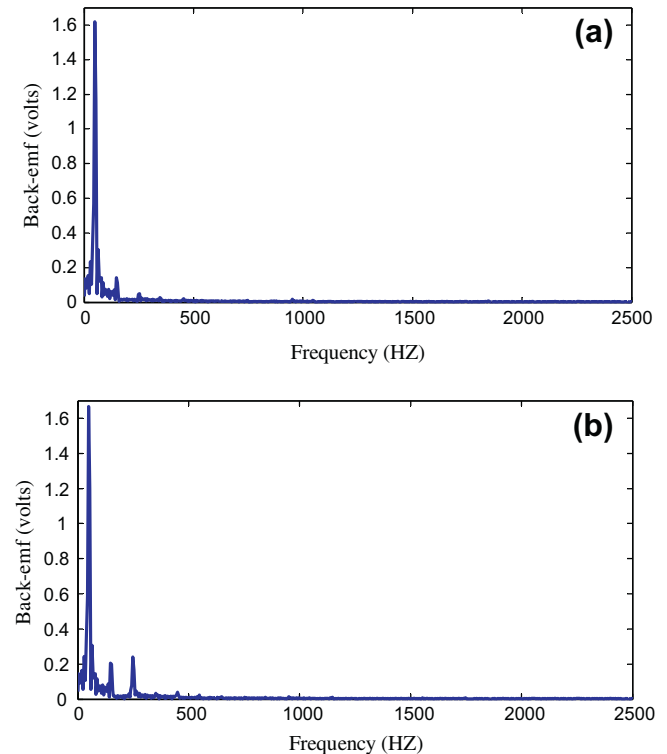


Fig. 13. The frequency spectrum of optimum back-emf wave forms are obtained from (a) the conventional skewing method and (b) the slot opening shift method.

Table 1
THD results.

	Conventional skewing method	Slot opening shift method
THD	0.0902	0.1931

5. Optimization process

In this section, our goal is finding the optimal value of the angular shift of the slot opening position of the first and third layers than the middle layer using the different algorithms, such as, RSM, GA and PSO.

5.1. Optimization using RSM

Response surface methodology is a collection of mathematical and statistical techniques that are useful for the modeling and analysis of problems in which an interest response is affected by several variables and the goal is to optimize this response [41]. Response surface can be presented as follows:

$$y = f(x_1, x_2, \dots, x_k) \quad (3)$$

where k is number of the independent variables.

However, above mentioned response surface is unknown in most of RSM problems. Thus, the first step in RSM is finding the suitable functional relationship between the response and independent variables. Usually, a low-order polynomial, such as a second-order polynomial is used as the approximation relationship between response and independent variables in some region of these variables. The general form of these second-order polynomial is as follows:

$$y = b_0 + \sum_{i=1}^k b_{ii}x_i + \sum_{i=1}^k b_{ii}x_i^2 + \sum_{i < j} b_{ij}x_i x_j + \epsilon \quad (4)$$

5.1.1. Objective function

We select the cogging torque as objective function in the optimization process using RSM. The objective function is defined as follows:

$$\text{Minimize } T_{\text{cog}} = f(\beta_1, \beta_2)$$

Including the following constraint⁴:

$$0^\circ \leq \beta_1, \beta_2 \leq 3.75^\circ$$

where β_1 and β_2 are the optimization parameters. These parameters show the angular shift of the first and third layers than the middle layer, respectively.

5.1.2. Optimization steps using RSM

The flowchart of optimization steps in response surface methodology is shown in Fig. 15.

First, the experiments are designed depend on number of factors, replicates, blocks and design type (central composite design). Then, these experiments are done using FEA of the PM-BLDC motor. After completing the experiments, the statistical analysis is done and its results are shown in Table 2. Table 2 shows the adequacy measures, such as, R^2 , adjusted R^2 , predicted R^2 and PRESS.

$$R^2 = \frac{SS_{\text{Model}}}{SS_{\text{Total}}} = 89.65\%$$

This value of R^2 shows that about 89.65% of the cogging torque variability expressed by two factors β_1 and β_2 . Therefore, the approximated model provides a good fit to the data.

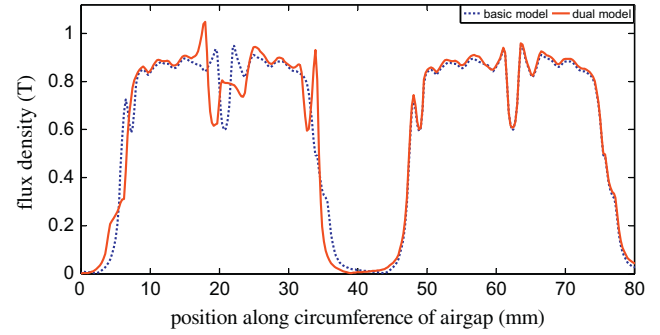


Fig. 14. The air-gap flux density distribution for basic and dual models.

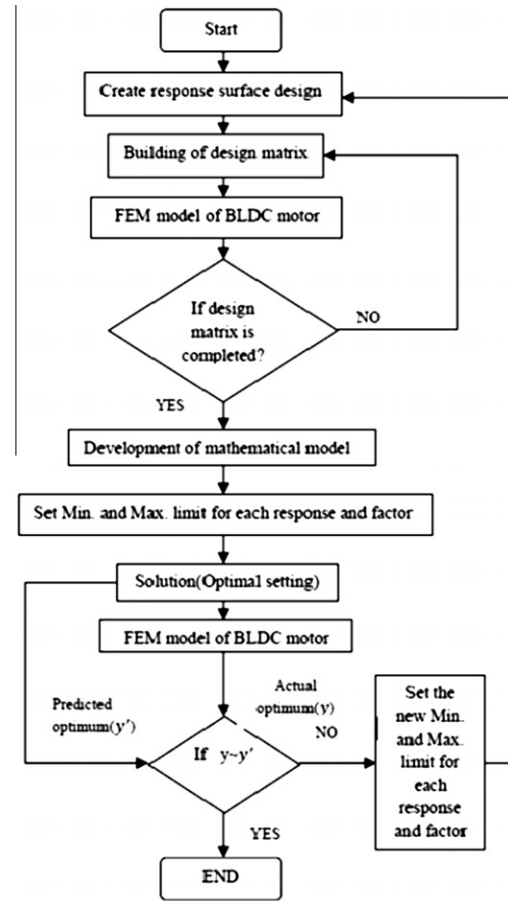


Fig. 15. Optimization steps [20].

Adjusted $R^2 = 78.74\%$

Adjusted R^2 is a statistic that is adjusted for the model size. The meaning of model size is the number of factors that represent the model. Adjusted R^2 will decrease if the insignificant factors are included in model.

PRESS : 0.8786

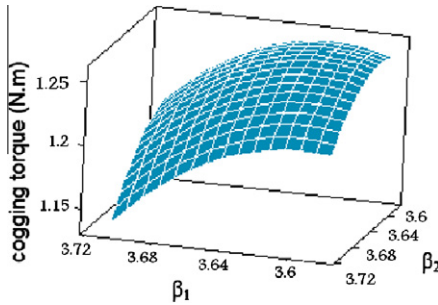
PRESS (prediction error sum of square) statistic is a measure that shows how well the model predicts the new data. A small value of PRESS indicates that model is likely a good predictor. The developed model in term of the uncoded factors obtained from RSM is shown in below:

⁴ 3.75° is the half of α that is introduced in Fig. 4.

Table 2

The statistical results using RSM for cogging torque.

Target term	Cogging torque		
	Coefficient	t-Test	p-Value
Const.	−3076.49	−1.669	0.039
β_1	1012.43	1.656	0.042
β_2	706.68	1.156	0.086
$\beta_1 * \beta_1$	−123.55	−1.781	0.118
$\beta_2 * \beta_2$	−81.5	−1.175	0.179
$\beta_1 * \beta_2$	−33.16	−0.352	0.035
R^2 : 89.65%		R^2 (adj): 78.74%	
R^2 (pred): 42.56%		PRESS: 0.8786	

**Fig. 16.** Response surface plot of cogging torque in term of the optimization parameters.

$$T_{cog} = -3076.49 + 1012.43\beta_1 + 706.68\beta_2 - 123.55(\beta_1)^2 - 81.5(\beta_2)^2 - 33.16(\beta_1 * \beta_2) \quad (5)$$

With respect to the statistical results of RSM, we conclude that the developed model can be used as the response surface in optimization process. In Fig. 16, the response surface plot of cogging torque versus the design variables is shown.

The optimization results using RSM is shown in Table 3⁵.

5.1.3. Validation of the developed model

In order to validate the developed model, the predicted result is compared with the FEA result. The comparison result is shown in Table 4.

The validation result express that the developed model is accurate sufficiently as the error percentage in approximation is small.

5.2. Optimization using GA and PSO

In order to optimization using GA and PSO, first, it is necessary that the objective function is specified. In this paper, the goal is the cogging torque reduction. Thus, to derive the objective function, first, we should extract an analytical model for cogging torque.

5.2.1. Analytical model for cogging torque with considering the slot opening shift

The cogging torque is defined as negative derivative of magnetic co-energy respect to the rotor position as follows [42]:

$$T_{cog} = -\frac{\partial W}{\partial \theta} \quad (6)$$

For simplification, we assume that the core iron is ideal, i.e. the stored magnetic energy in core is zero. Thus, the following relationship can be written for the magnetic co-energy [42]:

Table 3

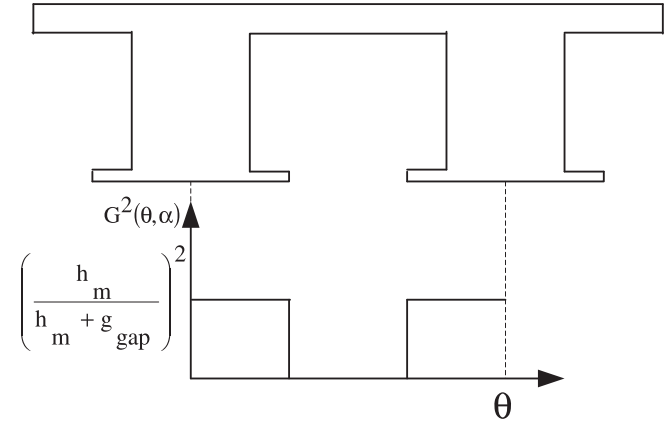
The optimization result using RSM.

β_1	β_2	Cogging torque (N m)
3.7068	3.7068	1.136

Table 4

Validation results.

β_1	β_2	FEA result (N m)	Predicted result (N m)	Error%
3.7068	3.7068	1.1485	1.136	1.25%

**Fig. 17.** The simplified model for square of the permeance function in a slot pitch.

$$\begin{aligned} Q &= W_{gap} + W_{PM} = \frac{1}{2\mu_0} \int_V B^2 dV \\ &= \frac{1}{2\mu_0} \int_V B_r^2(\theta) \left(\frac{h_m(\theta)}{h_m(\theta) + g(\theta, \alpha)} \right)^2 dV \rightarrow W \\ &= \frac{1}{2\mu_0} \int_V B_r^2(\theta) G^2(\theta, \alpha) dV \end{aligned} \quad (7)$$

where W is the stored magnetic co-energy in motor, α is rotor position, θ is angular position along circumference of air-gap, $B_r(\theta)$ is the flux density distribution of permanent magnet in along circumference of air-gap, $h_m(\theta)$ is the length of permanent magnet along the magnet magnetization and $g(\theta, \alpha)$ is the distribution of effective length of air-gap.

Since, we have accepted that the flux density distribution remain unchanged with affecting the slot opening shift, thus, the influence of slot opening shift is on the air-gap permeance function, only. According to above description, the following relationship can be written for $B_r^2(\theta)$ [42]:

$$B_r^2(\theta) = B_{r0}^2 + \sum_{n=1}^{\infty} B_{rn} \cos 2np\theta \quad (8)$$

where $B_{r0}^2 = B_r^2 \alpha_p$, $B_{rn} = \frac{2}{\pi n} B_r^2 \sin(n\alpha_p \pi)$, p is the number of pole pair, α_p is the ratio of pole arc to pole pitch, B_r is the residual flux density of permanent magnet.

With referring to [42], the cogging torque without skewing can be expressed as follows:

$$T_{cog}(\alpha) = \frac{\pi z L_{Fe}}{4\mu_0} (R_2^2 - R_1^2) \sum_{n=1}^{\infty} n G_n B_r \left(\frac{h_m}{g} \right) \sin(nz\alpha) \quad (9)$$

where, L_{Fe} is the axial length of armature, R_1 is the outer radius of armature, R_2 is the inner radius of stator yoke and z is number of the stator slots. As for Eq. (8) and with this assumption that the flux

⁵ In this paper, all of angles are in mechanical degree.

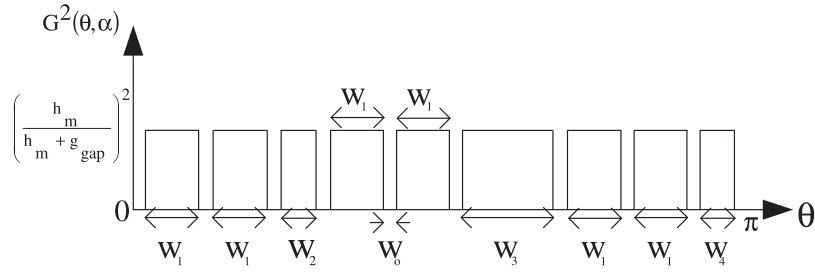


Fig. 18. The distribution of $G^2(\theta, \alpha)$.

Table 5
The optimization results using GA and PSO.

Harmonic number	GA			PSO		
	β_1	β_2	Cogging torque	β_1	β_2	Cogging torque
1	3.5308	3.71	1.237202	3.5542	3.7121	1.217535
2	3.5588	3.5559	1.295418	3.7100	3.5432	1.289637
3	3.7100	3.7100	1.146492	3.6866	3.6087	1.244461
4	3.5823	3.7100	1.216253	3.5566	3.6861	1.205539
5	3.5451	3.7100	1.232683	3.5617	3.7147	1.194315
6	3.5714	3.7100	1.224147	3.5192	3.7065	1.246057
7	3.6856	3.7100	1.197377	3.5502	3.7154	1.235707
8	3.5780	3.5530	1.287421	3.5010	3.5738	1.305987

Table 6
The best optimal result of the optimization process.

β_1	β_2	Cogging torque
3.71	3.71	1.146492

using GA and PSO, the optimal values of β_1 and β_2 are obtained as shown in Table 5⁶. In this table, the cogging torque is calculated using FEA via the optimal settings is obtained from GA and PSO.

With comparison of optimization results, it is concluded that the optimal results of RSM, GA and PSO have good agreement with each other. In order to comparison, the cogging torque wave form versus the rotor position in one period are illustrated in Fig. 19 for the basic model and the best optimal Setting that is obtained from GA, as follows (Table 6)⁷:

As shown in Fig. 19, the cogging torque reduce significantly about 77% by using slot opening shift.

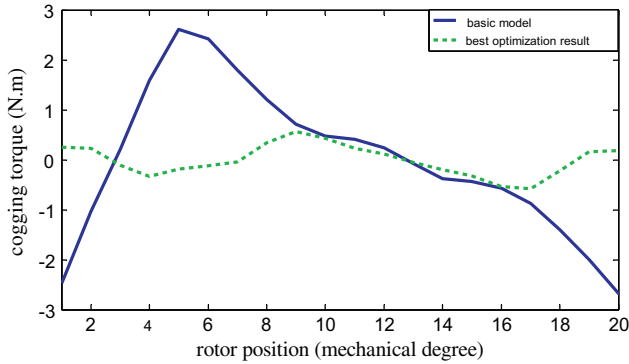


Fig. 19. The cogging torque comparison between the basic model and the best optimum result.

density distribution will not change, we conclude that the cogging torque is proportional the n th harmonic of the squared air-gap permeance function (G_n). G_n is defined as follows:

$$G_n = \frac{2}{\pi} \int_0^\pi G^2(\theta, \alpha) \cos(n\theta) d\theta \quad (10)$$

Therefore, we can minimize G_n for the entire harmonics to reduce the cogging torque. Consequently, all of components of G_n are selected as the objective function. We consider a simplified model without tooth-tip for the air-gap permeance function similar to Fig. 17.

By using this simplified model for our favorite motor, $G^2(\theta, \alpha)$ is obtained as show in Fig. 18. The non-uniform distribution of tooth width is obvious in this figure. In Fig. 18, W_0 is the slot opening width, W_1 is tooth width similar to the basic model, $W_2 = W_1 - \beta_1$, $W_3 = W_1 + \beta_1 + \beta_2$ and $W_4 = W_1 - \beta_2$. β_1 and β_2 are the angular shift of slot opening of the first and third layers in 3D slot opening skew model, respectively.

Therefore, with consideration all of harmonics of $G^2(\theta, \alpha)$ (G_n) as the objective function in the cogging torque optimization process

6. Conclusion

In this paper, the influence of skewing methods, i.e. the conventional skewing and the slot opening skew on the cogging torque reduction is investigated. For this purpose, we considered a three layer stator model for a typical six-pole PM-BLDC motor and the angular shifts of the first and third layers than the middle layer introduced as the optimization parameters. For simplifying in implementation of the slot opening skew model, a 2D dual model is presented instead of the 3D model. Generally, the following results can be drawn:

- The both skewing methods lead to the drastic reduction of cogging torque.
- The back-emf waveform is almost sinusoidal using the conventional skewing method.
- The trapezoidal waveform of back-emf is maintained nearly using the slot opening shift method.
- Since, the flux density distribution remain unchanged approximately using slot opening shift method, it is predicted that the average torque is not reduced by this method unlike other cogging torque reduction methods, such as, the magnet segmentation method. The cogging torque reduction is associated with the average torque reduction in the magnet segmentation method.
- Finally, the optimal settings resulting from optimization process show the major reduction in cogging torque about 77%.

⁶ In this table, the results are shown up to eighth harmonic. Also, all of angles are in mechanical degree and the cogging torque is in newton-meter.

⁷ It is noteworthy which there is very low difference between the optimal results are obtained from GA and RSM.

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